Prog4

```sql
CREATE TABLE employees (
    emp_id INT PRIMARY KEY,
    emp_name VARCHAR(30),
    department VARCHAR(20),
    city VARCHAR(20),
    salary INT,
    commission INT,
    join_date DATE );
INSERT INTO employees VALUES
(1, 'Anil',   'IT',      'Bengaluru', 50000, 5000, '2020-01-15'),
(2, 'Bhavna', 'HR',      'Mysuru',    40000, NULL, '2019-06-10'),
(3, 'Charan', 'IT',      'Mumbai',    60000, 8000, '2021-03-25'),
(4, 'Divya',  'Finance', 'Bengaluru', 45000, NULL, '2018-11-05'),
(5, 'Esha',   'Finance', 'Chennai',   55000, 3000, '2022-08-20');
CREATE TABLE contractors (
    con_id   INT PRIMARY KEY,
    con_name VARCHAR(30),
    city     VARCHAR(20)
);
INSERT INTO contractors VALUES
(1, 'Farhan', 'Mumbai'),
(2, 'Gita',   'Bengaluru'),
(3, 'Harsh',  'Delhi');
SELECT CAST('123' AS UNSIGNED) AS string_to_number;
SELECT CAST(salary AS CHAR) AS salary_as_text FROM employees;
SELECT CAST('2024-05-20' AS DATE) AS string_to_date;
SELECT CONVERT('456', UNSIGNED) AS string_to_number;
SELECT CONVERT(join_date, CHAR) AS date_as_text FROM employees;
SELECT STR_TO_DATE('20-05-2024', '%d-%m-%Y') AS converted_date;
SELECT emp_name, DATE_FORMAT(join_date, '%d-%m-%Y') AS formatted_date FROM employees;
SELECT NOW() AS current_datetime,
       CURDATE() AS current_date_only,
       CURTIME() AS current_time_only;
```

```sql
SELECT emp_name, DATE_FORMAT(join_date, '%d-%m-%Y') AS formatted_date FROM employees;
SELECT NOW() AS current_datetime,
       CURDATE() AS current_date_only,
       CURTIME() AS current_time_only;
SELECT emp_name,
       join_date,
       YEAR(join_date)AS join_year,
       MONTH(join_date)AS join_month,
       DAY(join_date)AS join_day,
       DAYNAME(join_date) AS day_name,
       MONTHNAME(join_date) AS month_name FROM employees;
SELECT emp_name,
       join_date,
       DATE_ADD(join_date, INTERVAL 10 DAY)AS plus_10_days,
       DATE_ADD(join_date, INTERVAL 3 MONTH)AS plus_3_months,
       DATE_SUB(join_date, INTERVAL 1 YEAR) AS minus_1_year FROM employees;
SELECT emp_name,
       DATEDIFF(CURDATE(), join_date) AS days_worked,
       TIMESTAMPDIFF(YEAR, join_date, CURDATE()) AS years_worked FROM employees;
SELECT emp_name, join_date, LAST_DAY(join_date) AS month_end FROM employees;
```

Output:

```
+------------------+
| string_to_number |
+------------------+
|              123 |
+------------------+
+----------------+
| salary_as_text |
+----------------+
| 50000          |
| 40000          |
| 60000          |
| 45000          |
| 55000          |
+----------------+
+----------------+
| string_to_date |
+----------------+
| 2024-05-20     |
+----------------+
+------------------+
| string_to_number |
+------------------+
|              456 |
+------------------+
```

```
+--------------+
| date_as_text |
+--------------+
| 2020-01-15   |
| 2019-06-10   |
| 2021-03-25   |
| 2018-11-05   |
| 2022-08-20   |
+--------------+
+----------------+
| converted_date |
+----------------+
| 2024-05-20     |
+----------------+
+----------+----------------+
| emp_name | formatted_date |
+----------+----------------+
| Anil     | 15-01-2020     |
| Bhavna   | 10-06-2019     |
| Charan   | 25-03-2021     |
| Divya    | 05-11-2018     |
| Esha     | 20-08-2022     |
+----------+----------------+
```

```
+---------------------+-------------------+--------------------+
| current_datetime    | current_date_only | current_time_only |
+---------------------+-------------------+--------------------+
| 2026-10-02 15:48:46 | 2026-10-02        | 15:48:46          |
+---------------------+-------------------+--------------------+
```

emp_name	join_date	join_year	join_month	join_day	day_name	month_name
Anil	2020-01-15	2020	1	15	Wednesday	January
Bhavna	2019-06-10	2019	6	10	Monday	June
Charan	2021-03-25	2021	3	25	Thursday	March
Divya	2018-11-05	2018	11	5	Monday	November
Esha	2022-08-20	2022	8	20	Saturday	August

emp_name	join_date	plus_10_days	plus_3_months	minus_1_year
Anil	2020-01-15	2020-01-25	2020-04-15	2019-01-15
Bhavna	2019-06-10	2019-06-20	2019-09-10	2018-06-10
Charan	2021-03-25	2021-04-04	2021-06-25	2020-03-25
Divya	2018-11-05	2018-11-15	2019-02-05	2017-11-05
Esha	2022-08-20	2022-08-30	2022-11-20	2021-08-20

emp_name	join_date	plus_10_days	plus_3_months	minus_1_year
Anil	2020-01-15	2020-01-25	2020-04-15	2019-01-15
Bhavna	2019-06-10	2019-06-20	2019-09-10	2018-06-10
Charan	2021-03-25	2021-04-04	2021-06-25	2020-03-25
Divya	2018-11-05	2018-11-15	2019-02-05	2017-11-05
Esha	2022-08-20	2022-08-30	2022-11-20	2021-08-20

emp_name	days_worked	years_worked
Anil	2452	6
Bhavna	2671	7
Charan	2017	5
Divya	2888	7
Esha	1504	4

emp_name	join_date	month_end
Anil	2020-01-15	2020-01-31
Bhavna	2019-06-10	2019-06-30
Charan	2021-03-25	2021-03-31
Divya	2018-11-05	2018-11-30
Esha	2022-08-20	2022-08-31

Prog5

```sql
CREATE TABLE employees (
    emp_id INT PRIMARY KEY,
    emp_name VARCHAR(30),
    department VARCHAR(20),
    city VARCHAR(20),
    salary INT,
    commission INT,
    join_date DATE
);
INSERT INTO employees VALUES
(1, 'Anil',   'IT',      'Bengaluru', 50000, 5000, '2020-01-15'),
(2, 'Bhavna', 'HR',      'Mysuru',    40000, NULL, '2019-06-10'),
(3, 'Charan', 'IT',      'Mumbai',    60000, 8000, '2021-03-25'),
(4, 'Divya',  'Finance', 'Bengaluru', 45000, NULL, '2018-11-05'),
(5, 'Esha',   'Finance', 'Chennai',   55000, 3000, '2022-08-20');
CREATE TABLE contractors (
    con_id   INT PRIMARY KEY,
    con_name VARCHAR(30),
    city     VARCHAR(20)
);
INSERT INTO contractors VALUES
(1, 'Farhan', 'Mumbai'),
(2, 'Gita',   'Bengaluru'),
(3, 'Harsh',  'Delhi');
SELECT 10 + 5  AS addition,
       10 - 5  AS subtraction,
       10 * 5  AS multiplication,
       10 / 4  AS division,
       10 DIV 4 AS integer_division,
       10 % 3  AS modulus;
SELECT emp_name,
       salary,
       salary + 5000        AS salary_with_bonus,
```

```
        10 / 4  AS division,
        10 DIV 4 AS integer_division,
        10 % 3  AS modulus;
SELECT emp_name,
        salary,
        salary + 5000        AS salary_with_bonus,
        salary * 12          AS yearly_salary,
        salary - (salary*0.1) AS salary_after_10pct_tax
FROM employees;
SELECT * FROM employees WHERE salary = 50000;
SELECT * FROM employees WHERE salary <> 50000;
SELECT * FROM employees WHERE salary > 45000;
SELECT * FROM employees WHERE salary < 50000;
SELECT * FROM employees WHERE salary >= 55000;
SELECT * FROM employees WHERE salary <= 45000;
SELECT * FROM employees
WHERE department = 'IT' AND salary > 50000;
SELECT * FROM employees
WHERE department = 'HR' OR department = 'Finance';
SELECT * FROM employees
WHERE NOT department = 'IT';
SELECT * FROM employees
WHERE (department = 'IT' OR department = 'Finance')
  AND salary >= 50000;
```

```
+----------+-------------+----------------+----------+------------------+---------+
| addition | subtraction | multiplication | division | integer_division | modulus |
+----------+-------------+----------------+----------+------------------+---------+
|       15 |           5 |             50 |   2.5000 |                2 |       1 |
+----------+-------------+----------------+----------+------------------+---------+
```

```
+----------+--------+-------------------+---------------+------------------------+
| emp_name | salary | salary_with_bonus | yearly_salary | salary_after_10pct_tax |
+----------+--------+-------------------+---------------+------------------------+
| Anil     |  50000 |             55000 |        600000 |                45000.0 |
| Bhavna   |  40000 |             45000 |        480000 |                36000.0 |
| Charan   |  60000 |             65000 |        720000 |                54000.0 |
| Divya    |  45000 |             50000 |        540000 |                40500.0 |
| Esha     |  55000 |             60000 |        660000 |                49500.0 |
+----------+--------+-------------------+---------------+------------------------+
```

```
+--------+----------+------------+-----------+--------+------------+------------+
| emp_id | emp_name | department | city      | salary | commission | join_date  |
+--------+----------+------------+-----------+--------+------------+------------+
|      1 | Anil     | IT         | Bengaluru |  50000 |       5000 | 2020-01-15 |
+--------+----------+------------+-----------+--------+------------+------------+
```

```
+--------+----------+------------+-----------+--------+------------+------------+
| emp_id | emp_name | department | city      | salary | commission | join_date  |
+--------+----------+------------+-----------+--------+------------+------------+
|      2 | Bhavna   | HR         | Mysuru    |  40000 |       NULL | 2019-06-10 |
|      3 | Charan   | IT         | Mumbai    |  60000 |       8000 | 2021-03-25 |
|      4 | Divya    | Finance    | Bengaluru |  45000 |       NULL | 2018-11-05 |
|      5 | Esha     | Finance    | Chennai   |  55000 |       3000 | 2022-08-20 |
+--------+----------+------------+-----------+--------+------------+------------+
```

```
+--------+----------+------------+-----------+--------+------------+------------+
| emp_id | emp_name | department | city      | salary | commission | join_date  |
+--------+----------+------------+-----------+--------+------------+------------+
|      1 | Anil     | IT         | Bengaluru |  50000 |       5000 | 2020-01-15 |
|      3 | Charan   | IT         | Mumbai    |  60000 |       8000 | 2021-03-25 |
|      5 | Esha     | Finance    | Chennai   |  55000 |       3000 | 2022-08-20 |
+--------+----------+------------+-----------+--------+------------+------------+
```

```
+--------+----------+------------+-----------+--------+------------+------------+
| emp_id | emp_name | department | city      | salary | commission | join_date  |
+--------+----------+------------+-----------+--------+------------+------------+
|      2 | Bhavna   | HR         | Mysuru    |  40000 |       NULL | 2019-06-10 |
|      4 | Divya    | Finance    | Bengaluru |  45000 |       NULL | 2018-11-05 |
+--------+----------+------------+-----------+--------+------------+------------+
```

```
+--------+----------+------------+---------+--------+------------+------------+
| emp_id | emp_name | department | city    | salary | commission | join_date  |
+--------+----------+------------+---------+--------+------------+------------+
|      3 | Charan   | IT         | Mumbai  |  60000 |       8000 | 2021-03-25 |
|      5 | Esha     | Finance    | Chennai |  55000 |       3000 | 2022-08-20 |
+--------+----------+------------+---------+--------+------------+------------+
```

```
+--------+----------+------------+-----------+--------+------------+------------+
| emp_id | emp_name | department | city      | salary | commission | join_date  |
+--------+----------+------------+-----------+--------+------------+------------+
|      2 | Bhavna   | HR         | Mysuru    |  40000 |       NULL | 2019-06-10 |
|      4 | Divya    | Finance    | Bengaluru |  45000 |       NULL | 2018-11-05 |
+--------+----------+------------+-----------+--------+------------+------------+
+--------+----------+------------+--------+--------+------------+------------+
| emp_id | emp_name | department | city   | salary | commission | join_date  |
+--------+----------+------------+--------+--------+------------+------------+
|      3 | Charan   | IT         | Mumbai |  60000 |       8000 | 2021-03-25 |
+--------+----------+------------+--------+--------+------------+------------+
+--------+----------+------------+-----------+--------+------------+------------+
| emp_id | emp_name | department | city      | salary | commission | join_date  |
+--------+----------+------------+-----------+--------+------------+------------+
|      2 | Bhavna   | HR         | Mysuru    |  40000 |       NULL | 2019-06-10 |
|      4 | Divya    | Finance    | Bengaluru |  45000 |       NULL | 2018-11-05 |
|      5 | Esha     | Finance    | Chennai   |  55000 |       3000 | 2022-08-20 |
+--------+----------+------------+-----------+--------+------------+------------+
+--------+----------+------------+-----------+--------+------------+------------+
| emp_id | emp_name | department | city      | salary | commission | join_date  |
+--------+----------+------------+-----------+--------+------------+------------+
|      2 | Bhavna   | HR         | Mysuru    |  40000 |       NULL | 2019-06-10 |
|      4 | Divya    | Finance    | Bengaluru |  45000 |       NULL | 2018-11-05 |
|      5 | Esha     | Finance    | Chennai   |  55000 |       3000 | 2022-08-20 |
+--------+----------+------------+-----------+--------+------------+------------+
```

```
+--------+----------+------------+--------+--------+------------+------------+
| emp_id | emp_name | department | city   | salary | commission | join_date  |
+--------+----------+------------+--------+--------+------------+------------+
|      3 | Charan   | IT         | Mumbai |  60000 |       8000 | 2021-03-25 |
+--------+----------+------------+--------+--------+------------+------------+
```

```
+--------+----------+------------+----------+--------+------------+------------+
| emp_id | emp_name | department | city     | salary | commission | join_date  |
+--------+----------+------------+----------+--------+------------+------------+
|      2 | Bhavna   | HR         | Mysuru   |  40000 |       NULL | 2019-06-10 |
|      4 | Divya    | Finance    | Bengaluru |  45000 |       NULL | 2018-11-05 |
|      5 | Esha     | Finance    | Chennai  |  55000 |       3000 | 2022-08-20 |
+--------+----------+------------+----------+--------+------------+------------+
```


```
+--------+----------+------------+----------+--------+------------+------------+
| emp_id | emp_name | department | city     | salary | commission | join_date  |
+--------+----------+------------+----------+--------+------------+------------+
|      2 | Bhavna   | HR         | Mysuru   |  40000 |       NULL | 2019-06-10 |
|      4 | Divya    | Finance    | Bengaluru |  45000 |       NULL | 2018-11-05 |
|      5 | Esha     | Finance    | Chennai  |  55000 |       3000 | 2022-08-20 |
+--------+----------+------------+----------+--------+------------+------------+
```


```
+--------+----------+------------+----------+--------+------------+------------+
| emp_id | emp_name | department | city     | salary | commission | join_date  |
+--------+----------+------------+----------+--------+------------+------------+
|      1 | Anil     | IT         | Bengaluru |  50000 |       5000 | 2020-01-15 |
|      3 | Charan   | IT         | Mumbai   |  60000 |       8000 | 2021-03-25 |
|      5 | Esha     | Finance    | Chennai  |  55000 |       3000 | 2022-08-20 |
+--------+----------+------------+----------+--------+------------+------------+
```

Prog6

```sql
CREATE TABLE employees (
    emp_id INT PRIMARY KEY,
    emp_name VARCHAR(30),
    department VARCHAR(20),
    city VARCHAR(20),
    salary INT,
    commission INT,
    join_date DATE);
INSERT INTO employees VALUES
(1, 'Anil',   'IT',      'Bengaluru', 50000, 5000, '2020-01-15'),
(2, 'Bhavna', 'HR',      'Mysuru',    40000, NULL, '2019-06-10'),
(3, 'Charan', 'IT',      'Mumbai',    60000, 8000, '2021-03-25'),
(4, 'Divya',  'Finance', 'Bengaluru', 45000, NULL, '2018-11-05'),
(5, 'Esha',   'Finance', 'Chennai',   55000, 3000, '2022-08-20');
CREATE TABLE contractors (
    con_id   INT PRIMARY KEY,con_name VARCHAR(30),city     VARCHAR(20));
INSERT INTO contractors VALUES (1, 'Farhan', 'Mumbai'),(2, 'Gita',   'Bengaluru'),(3, 'Harsh',  'Delhi');
SELECT * FROM employees WHERE salary BETWEEN 45000 AND 55000;
SELECT * FROM employees WHERE salary NOT BETWEEN 45000 AND 55000;
SELECT * FROM employees WHERE city IN ('Bengaluru', 'Mumbai');
SELECT * FROM employees WHERE city NOT IN ('Bengaluru', 'Mumbai');
SELECT * FROM employees WHERE emp_name LIKE 'A%';
SELECT * FROM employees WHERE emp_name LIKE '%a';
SELECT * FROM employees WHERE emp_name LIKE '_h%';
SELECT * FROM employees WHERE commission IS NULL;
SELECT * FROM employees WHERE commission IS NOT NULL;
SELECT emp_name FROM employees e
WHERE EXISTS (SELECT 1 FROM contractors c WHERE c.city = e.city);
SELECT * FROM employees
WHERE salary > ALL (SELECT salary FROM employees WHERE department = 'HR');
SELECT city FROM employees UNION SELECT city FROM contractors;
SELECT city FROM employees UNION ALL SELECT city FROM contractors;
SELECT emp_name AS name, city, 'Employee' AS type FROM employees
UNION SELECT con_name, city, 'Contractor' FROM contractors;
```

```
+--------+----------+------------+-----------+--------+------------+------------+
| emp_id | emp_name | department | city      | salary | commission | join_date  |
+--------+----------+------------+-----------+--------+------------+------------+
|      1 | Anil     | IT         | Bengaluru |  50000 |       5000 | 2020-01-15 |
|      4 | Divya    | Finance    | Bengaluru |  45000 |       NULL | 2018-11-05 |
|      5 | Esha     | Finance    | Chennai   |  55000 |       3000 | 2022-08-20 |
+--------+----------+------------+-----------+--------+------------+------------+
+--------+----------+------------+--------+--------+------------+------------+
| emp_id | emp_name | department | city   | salary | commission | join_date  |
+--------+----------+------------+--------+--------+------------+------------+
|      2 | Bhavna   | HR         | Mysuru |  40000 |       NULL | 2019-06-10 |
|      3 | Charan   | IT         | Mumbai |  60000 |       8000 | 2021-03-25 |
+--------+----------+------------+--------+--------+------------+------------+
+--------+----------+------------+-----------+--------+------------+------------+
| emp_id | emp_name | department | city      | salary | commission | join_date  |
+--------+----------+------------+-----------+--------+------------+------------+
|      1 | Anil     | IT         | Bengaluru |  50000 |       5000 | 2020-01-15 |
|      3 | Charan   | IT         | Mumbai    |  60000 |       8000 | 2021-03-25 |
|      4 | Divya    | Finance    | Bengaluru |  45000 |       NULL | 2018-11-05 |
+--------+----------+------------+-----------+--------+------------+------------+
+--------+----------+------------+---------+--------+------------+------------+
| emp_id | emp_name | department | city    | salary | commission | join_date  |
+--------+----------+------------+---------+--------+------------+------------+
|      2 | Bhavna   | HR         | Mysuru  |  40000 |       NULL | 2019-06-10 |
|      5 | Esha     | Finance    | Chennai |  55000 |       3000 | 2022-08-20 |
+--------+----------+------------+---------+--------+------------+------------+
```

```
+--------+----------+------------+-----------+--------+------------+------------+
| emp_id | emp_name | department | city      | salary | commission | join_date  |
+--------+----------+------------+-----------+--------+------------+------------+
|      1 | Anil     | IT         | Bengaluru |  50000 |       5000 | 2020-01-15 |
+--------+----------+------------+-----------+--------+------------+------------+
```

```
+--------+----------+------------+-----------+--------+------------+------------+
| emp_id | emp_name | department | city      | salary | commission | join_date  |
+--------+----------+------------+-----------+--------+------------+------------+
|      2 | Bhavna   | HR         | Mysuru    |  40000 |       NULL | 2019-06-10 |
|      4 | Divya    | Finance    | Bengaluru |  45000 |       NULL | 2018-11-05 |
|      5 | Esha     | Finance    | Chennai   |  55000 |       3000 | 2022-08-20 |
+--------+----------+------------+-----------+--------+------------+------------+
```

```
+--------+----------+------------+--------+--------+------------+------------+
| emp_id | emp_name | department | city   | salary | commission | join_date  |
+--------+----------+------------+--------+--------+------------+------------+
|      2 | Bhavna   | HR         | Mysuru |  40000 |       NULL | 2019-06-10 |
|      3 | Charan   | IT         | Mumbai |  60000 |       8000 | 2021-03-25 |
+--------+----------+------------+--------+--------+------------+------------+
```

```
+--------+----------+------------+-----------+--------+------------+------------+
| emp_id | emp_name | department | city      | salary | commission | join_date  |
+--------+----------+------------+-----------+--------+------------+------------+
|      2 | Bhavna   | HR         | Mysuru    |  40000 |       NULL | 2019-06-10 |
|      4 | Divya    | Finance    | Bengaluru |  45000 |       NULL | 2018-11-05 |
+--------+----------+------------+-----------+--------+------------+------------+
```

```
+--------+----------+------------+-----------+--------+------------+------------+
| emp_id | emp_name | department | city      | salary | commission | join_date  |
+--------+----------+------------+-----------+--------+------------+------------+
|      1 | Anil     | IT         | Bengaluru |  50000 |       5000 | 2020-01-15 |
|      3 | Charan   | IT         | Mumbai    |  60000 |       8000 | 2021-03-25 |
|      5 | Esha     | Finance    | Chennai   |  55000 |       3000 | 2022-08-20 |
+--------+----------+------------+-----------+--------+------------+------------+
```

```
+----------+
| emp_name |
+----------+
| Anil     |
| Charan   |
| Divya    |
+----------+
```

```
+--------+----------+------------+-----------+--------+------------+------------+
| emp_id | emp_name | department | city      | salary | commission | join_date  |
+--------+----------+------------+-----------+--------+------------+------------+
|      1 | Anil     | IT         | Bengaluru |  50000 |       5000 | 2020-01-15 |
|      3 | Charan   | IT         | Mumbai    |  60000 |       8000 | 2021-03-25 |
|      4 | Divya    | Finance    | Bengaluru |  45000 |       NULL | 2018-11-05 |
|      5 | Esha     | Finance    | Chennai   |  55000 |       3000 | 2022-08-20 |
+--------+----------+------------+-----------+--------+------------+------------+
```

```
+-----------+
| city      |
+-----------+
| Bengaluru |
| Mysuru    |
| Mumbai    |
| Chennai   |
| Delhi     |
+-----------+
```

```
+-----------+
| city      |
+-----------+
| Bengaluru |
| Mysuru    |
| Mumbai    |
| Bengaluru |
| Chennai   |
| Mumbai    |
| Bengaluru |
| Delhi     |
+-----------+
```

```
+--------+-----------+------------+
| name   | city      | type       |
+--------+-----------+------------+
| Anil   | Bengaluru | Employee   |
| Bhavna | Mysuru    | Employee   |
| Charan | Mumbai    | Employee   |
```